

Dmitriy Tychshenko

Principal Engineer · Full Stack Tech Lead · AI Product Engineering

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SUMMARY

Principal-level full stack engineer and tech lead with 15+ years of building large-scale, high-performance web applications for enterprise, financial and consumer platforms. Deep expertise in architecture, performance optimization and developer productivity, focused on React, Node.js and TypeScript. Now bringing AI into products, with hands-on experience in LLM function calling, the Model Context Protocol (MCP) and retrieval-augmented generation (RAG), including leading an AI-native product prototype. Loves diving into complex problems, from refactoring legacy code to architecting micro-frontend systems that actually scale, and taking products from idea to production. Founder of Doge Consulting Inc. (est. 2015).

Launch lead
TurboTax Live Full Service in Canada, where a dedicated tax expert handles the whole return

1st
Fully online T2 corporate tax filing in Canada. Key role in the launch, led its server-side rendering

60K+
Peak concurrent users on TurboTax Online, shipped across the full stack

80%+
Faster response times after rebuilding a tax estimation API

30s → 2–3s
Developer experience win: local feedback loop for engineers building Citi's global trading platform

CORE SKILLS

AI & LLMs	Function calling (tool use) · Model Context Protocol (MCP) · retrieval-augmented generation (RAG) · vector search with Chroma DB · AI integration · AI-native product prototyping
Front end	React · TypeScript · Next.js · Vite · Mantine · Angular · NgRx · Redux · RxJS · Electron · Web Components · ag-Grid · server-side rendering
Back end	Node.js · Fastify · Java · Kotlin · Spring Boot · RxJava · GraphQL · REST APIs
Cloud & DevOps	GCP · AWS · GitHub Actions · Jenkins · CI/CD
Architecture	Micro-frontends · desktop interoperability (interop.io io.Connect) · NX monorepos · platform capabilities · performance profiling · automated end-to-end and unit testing · developer experience
Leadership	Technical scoping and estimation · coding standards and release checklists · code review and mentorship · partnering with product and design

EXPERIENCE

Senior Full Stack Engineer (Contract) | **TD Securities** — Jan 2026 – Present · Toronto, ON

FICC Unified Desktop: TD Securities' interop-based trading platform, hosting apps and automated workflows for front-office and operations users across fixed income, FX and commodities.

- Launched Atomic, a line of next-generation apps and MicroUIs that lift core workflows out of book-of-record platforms and into a workspace traders can shape themselves.
- Delivering trade automation across multiple desks, collaborative auction workflows and real-time inventory management for Global Precious Metals.
- Working across front end, back end, infrastructure, AI integration, DevOps and developer productivity.

Tech stack: React, Vite, Mantine, Node.js, Fastify, Java, Spring Boot, GitHub Actions, GCP, interop.io io.Connect

Principal Engineer | **Confidential AI-native startup (under NDA)** — Aug 2025 – Oct 2025 · Toronto, ON

Commercial real estate startup building a platform for short-term industrial and commercial space.

- Led development of the company's AI-native product prototype, built on Anthropic Claude models.
- Implemented function calling so the model can take actions inside the application.
- Built retrieval-augmented generation (RAG) with Chroma DB as the vector store.

Full Stack Engineering Lead (Contract) | **Intuit, TurboTax Online Canada** — Jan 2024 – Jun 2025 · Toronto, ON

- Rejoined to lead a core redesign of TurboTax navigation on a hub-and-spoke model, with dynamic guidance and context-aware Next Best Action suggestions that improved usability and task completion for filers.
- Played a key role in launching TurboTax Online Canada for Small Business, the first fully online solution for Canadian small business owners to file T2 corporate tax returns.
- Led the server-side rendering integration, improving performance, SEO and load times during peak filing season.

- Partnered with product, design and engineering leadership on scope and deliverables, and turned requirements into prioritized, estimated work for the team.
- Raised engineering quality with coding standards, release checklists, mentorship and code reviews.
- Built platform-level navigation capabilities that keep complex product surfaces consistent and flexible.

Tech stack: React, TypeScript, Node.js, Java, Kotlin, Spring, GraphQL, Jenkins, AWS

Senior Full Stack Engineer (Contract) | Citi

Jan 2023 – Jan 2024 · Mississauga, ON

- Rejoined to optimize and scale the global trading platform, improving performance and reliability by tuning streaming data updates, hardening state restoration for complex saved layouts and profiling rendering bottlenecks.
- Improved developer experience by enabling local overrides of individual micro-frontends, cutting feedback loops from about 30 seconds to 2–3 seconds, close to hot reload.
- Maintained and evolved the NX-based modular architecture for efficient cross-team development, and led core library and framework upgrades that closed security vulnerabilities.

Tech stack: Angular, TypeScript, Electron, NX, ag-Grid, RxJS, NgRx, Web Components

Senior Full Stack Engineer (Contract) | Intuit, TurboTax Canada and US

Oct 2019 – Jan 2023 · Mississauga, ON

- Rejoined to lead the launch of TurboTax Live Full Service in Canada, where a dedicated tax expert handles the return from start to finish through live chat, document upload, ID verification and digital signing.
- Designed and implemented a business model that matches each customer with the right product by tax complexity.
- Improved data import from financial institutions for TurboTax US, reducing manual entry and improving accuracy.
- Led the technical implementation of a redesigned onboarding experience, unified self-help across TurboTax Canada products, and contributed to a shared component runtime for the micro-frontend platform.

Senior Consultant (Contract) | Avanade

Mar 2018 – Oct 2019 · Mississauga and Toronto, ON

Engagements at Citi (Jul 2018 – Oct 2019) and RBC Capital Markets (Mar – Jun 2018).

- **Citi:** Engineered a high-performance $4B global trading platform that gives rates traders real-time financial data in customizable, tabbed workspaces.
- **Citi:** Built the Electron shell that hosts trading tools, persists user state and window layouts, and coordinates communication between windows, laying the foundation for the platform's multi-app ecosystem.
- **Citi:** Standardized state management with NgRx, introduced lazy-loaded routing that reduced startup time, and built reusable Angular component libraries (ng-packagr, NX) adopted across teams.
- **RBC Capital Markets:** Built a modular, responsive sales platform with a 360-degree client view and independently deployed plug-in components, using Material Design, mock backend services and 90%+ test coverage.

Senior Full Stack Engineer (Contract) | Intuit, TurboTax

Oct 2015 – Feb 2018 · Mississauga, ON

- Delivered full-stack improvements to TurboTax, a distributed platform serving 60,000+ concurrent users at peak.
- Migrated a tax estimation API from a legacy PHP service to Java RESTEasy, cutting average response times by over 80% with full unit and integration test coverage.
- Built an end-to-end testing framework with Nightwatch.js, reducing manual QA and raising release confidence.
- Won People's Choice at an Intuit hackathon with a speech recognition and live video tax prep prototype.

Tech stack: Java, Spring Boot, RxJava, Hibernate, Node.js, React, Redux, TypeScript

Consultant | Infusion

Jul 2011 – Oct 2015 · Toronto, ON

Clients in banking, insurance, financial services and consumer devices.

- Led front-end development of a front-line banking portal used by 40,000+ employees at a Big Five Canadian bank.
- Rebuilt an app store-style business intelligence platform for a Big Five Canadian bank, cutting load times by 60% on desktop and 80% on mobile, and cut page load times by 80% on a major bank's call center application.
- Built two working mobile banking prototypes for phone and tablet in 30 hours, helping retain a major client.
- Delivered a multilingual self-service support site with custom Lucene search for a global e-reader brand.
- Led on-site UAT sessions abroad on short notice, resolving a month's worth of defects in one week.
- Won Infusion's Global Code Complete Award for a project-tracking platform built for a global financial institution.

Earlier Career

2004 – 2011

Senior Support Analyst, KnowledgeWorx: helped build an ASP.NET and SharePoint monitoring app and cut QA time by 30% with test scripts. Earlier roles in project accounting at RWD Technologies and technical support at Towson University.

EDUCATION

Computer Programmer Diploma | Seneca College, Toronto, ON, Canada — 2009 – 2011

Bachelor of Science | Towson University, Towson, MD, USA — 2003 – 2007